

Exercise 4.3: Consider the problem,

$$\begin{aligned} &\text{minimize } f(x) = -x_1 - x_2 \\ &\text{subject to } x_1 + x_2 \leq 2 \\ &\quad x_1, x_2 \geq 0 \end{aligned}$$

1. Consider the feasible directions at $x = (0, 0)^T, (0, 1)^T, (1, 1)^T$, and $(0, 2)^T$.

Solution. Note that $(0, 0)^T$ only makes $x_1, x_2 \geq 0$ active. Therefore the feasible direction p is defined by,

$$(1, 0)(p_1, p_2)^T = p_1 \geq 0.$$

$$(0, 1)(p_1, p_2)^T = p_2 \geq 0.$$

Note that $(0, 1)^T$ only makes $x_1 \geq 0$ active. Therefore the feasible direction p is defined by,

$$(1, 0)(p_1, p_2)^T = p_1 \geq 0.$$

With $p_2 \in \mathbb{R}$. Note that $(1, 1)^T$ only makes $x_1 + x_2 \leq 2$. Therefore the feasible direction p is defined by,

$$(-1, -1)(p_1, p_2)^T = -p_1 - p_2 \geq 0.$$

Finally note that $(0, 2)^T$ makes $x_1 + x_2 \leq 2$ and $x_1 \geq 0$ active. Therefore the feasible direction p is defined by,

$$(-1, -1)(p_1, p_2)^T = -p_1 - p_2 \geq 0.$$

$$(1, 0)(p_1, p_2)^T = p_1 \geq 0.$$

□

2. Determine whether there exists feasible descent direction at these points, and hence determine which (if any) of the points can be local minimizers.

Solution. Recall that any direction p which satisfies $\nabla f(x)^T p < 0$, will be a descent direction. Since $f(x)$ is linear know that $\nabla f(x)^T = (-1, -1)$. Computing the inner product with p we get that,

$$(-1, -1)(p_1, p_2)^T = -p_1 - p_2 < 0.$$

Or rephrased $p_1 + p_2 > 0$. For point $(0, 0)$ with constraints $p_1, p_2 \geq 0$ every non-trivial feasible direction is a descent direction. For point $(0, 1)$ with constraint $p_1 \geq 0$ as long as $p_1 > p_2$ our feasible direction is a descent direction. For point $(1, 1)$ with constraint $p_1 + p_2 \leq 0$ every feasible direction is an ascent direction. For point $(0, 2)$ with constraint $p_1 + p_2 \leq 0$ and $p_1 \geq 0$ every feasible direction is an ascent direction. □

Exercise 6.5: Find the first three terms of the Taylor series for,

$$f(x_1, x_2) = \sqrt{x_1^2 + x_2^2}$$

about the point $x_0 = (3, 4)^T$.

Solution. Recall the Multidimensional Taylor Series expansion with cubic remainder,

$$f(x_0 + p) = f(x_0) + p^T \nabla f(x_0) + \frac{1}{2} p^T \nabla^2 f(x_0) p + g(\Xi)$$

Where g is some function cubic in p with a tensor shaped $\nabla^3 f(x_0)$ and $x_0 < \Xi < p$ (Book does not go into detail about how this term is notated). Computing $\nabla f(x)$ and $\nabla^2 f(x)$ we get,

$$\nabla f(x) = \begin{pmatrix} \frac{x}{\sqrt{x^2 + y^2}} \\ \frac{y}{\sqrt{x^2 + y^2}} \end{pmatrix},$$

$$\nabla^2 f(x) = \begin{pmatrix} \frac{y^2}{(x^2 + y^2) \sqrt{x^2 + y^2}} & -\frac{xy}{(x^2 + y^2)^{\frac{3}{2}}} \\ -\frac{xy}{(x^2 + y^2)^{\frac{3}{2}}} & \frac{x^2}{(x^2 + y^2) \sqrt{x^2 + y^2}} \end{pmatrix}.$$

Plugging in our $x_0 = (3, 4)^T$, we get

$$\nabla f(x_0) = \begin{pmatrix} \frac{3}{5} \\ \frac{4}{5} \end{pmatrix},$$

$$\nabla^2 f(x_0) = \begin{pmatrix} \frac{16}{125} & -\frac{12}{125} \\ -\frac{12}{125} & \frac{9}{125} \end{pmatrix}.$$

Finally for some $p = (p_1, p_2)^T$ we get the first three terms of the Taylor series are,

$$f(x_0 + p) \approx 5 + p^T \begin{pmatrix} \frac{3}{5} \\ \frac{4}{5} \end{pmatrix} + \frac{1}{2} p^T \begin{pmatrix} \frac{16}{125} & -\frac{12}{125} \\ -\frac{12}{125} & \frac{9}{125} \end{pmatrix} p$$

□

Exercise 1.1: Find the set of all feasible directions at points $x_a = (0, 0, 2)^T$, $x_b = (3, 0, 1)^T$, and $x_c = (1, 1, 1)^T$ for Example 3.1. The set of constraints from Example 3.1 are,

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 6, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Solution. Recall that p is a feasible direction with respect to the feasible set at a point x if and only if

$$\begin{aligned} a_i^T p &= 0, i \in \varepsilon \\ a_i^T p &\geq 0, i \in \hat{\tau} \end{aligned}$$

Where ε is the set of equality constraints, $\hat{\tau}$ is the set of active inequality constraints, and a_i are the vectors which define our constraints. For our set of constraints, $a_1 = (1, 2, 3)^T$, $a_2 = (1, 0, 0)^T$, $a_3 = (0, 1, 0)^T$, $a_4 = (0, 0, 1)^T$.

Consider point $x_a = (0, 0, 2)^T$. Note that the only active inequality constraints are a_2 , and a_3 . Therefore the following inequalities describe the set of all feasible directions,

$$a_1^T p = 1p_1 + 2p_2 + 3p_3 = 0$$

and

$$\begin{aligned} a_2^T p &= 1p_1 + 0p_2 + 0p_3 = p_1 \geq 0 \\ a_3^T p &= 0p_1 + 1p_2 + 0p_3 = p_2 \geq 0. \end{aligned}$$

Consider the point $x_b = (3, 0, 1)^T$. Note that the only active inequality constraint is a_3 . Therefore the following equalities and inequalities describe the set of all feasible directions,

$$a_1^T p = 1p_1 + 2p_2 + 3p_3 = 0$$

and

$$a_3^T p = 0p_1 + 1p_2 + 0p_3 = p_2 \geq 0.$$

Consider the point $x_c = (1, 1, 1)^T$. Note that there are no active inequality constraints so we only have to satisfy the equality constraint,

$$a_1^T p = 1p_1 + 2p_2 + 3p_3 = 0$$

□

Exercise 1.3: Consider the system of inequality constraints $Ax \leq b$ with

$$A = \begin{bmatrix} 9 & 4 & 1 & 9 & -7 \\ 6 & -7 & 8 & -4 & -6 \\ 1 & 6 & 3 & -7 & 6 \end{bmatrix} \text{ and } b = \begin{bmatrix} -15 \\ -30 \\ -20 \end{bmatrix}$$

For the given values of x and p , perform a ratio test to determine the maximum step length $\bar{\alpha}$ such that $x + \bar{\alpha}p$ remains feasible.

(i) $x = (8, 4, -3, 4, 1)^T$ and $p = (1, 1, 1, 1, 1)^T$

Solution. Evaluating Ax to see which constraints are active/inactive,

$$\begin{bmatrix} 9 & 4 & 1 & 9 & -7 \\ 6 & -7 & 8 & -4 & -6 \\ 1 & 6 & 3 & -7 & 6 \end{bmatrix} \begin{pmatrix} 8 \\ 4 \\ -3 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 114 \\ -26 \\ 1 \end{pmatrix}$$

Since each component of our result is strictly greater than b every inequality constraint is inactive.

Recall that by the ratio test we know that,

$$\bar{\alpha} = \min\{(a_i^T x - b_i)/(a_i^T p) : (a_i^T p) < 0\}$$

Computing all $(a_i^T p)$ or Ap to see which constraints bound our step,

$$\begin{bmatrix} 9 & 4 & 1 & 9 & -7 \\ 6 & -7 & 8 & -4 & -6 \\ 1 & 6 & 3 & -7 & 6 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 16 \\ -3 \\ 9 \end{pmatrix}$$

Since a_2 is the only constraint which our direction is set to intersect, it defines the $\bar{\alpha}$ in the ratio test,

$$\bar{\alpha} = ((6, -7, 8, -4, -6)(8, 4, -3, 4, 1)^T + 30)/3 = \frac{4}{3}.$$

□

(ii) $x = (7, -4, -3, -3, 3)^T$ and $p = (3, 2, 0, 1, -2)^T$

Solution. Evaluating Ax to see which constraints are active/inactive,

$$\begin{bmatrix} 9 & 4 & 1 & 9 & -7 \\ 6 & -7 & 8 & -4 & -6 \\ 1 & 6 & 3 & -7 & 6 \end{bmatrix} \begin{pmatrix} 7 \\ -4 \\ -3 \\ -3 \\ 3 \end{pmatrix} = \begin{pmatrix} -4 \\ 40 \\ 13 \end{pmatrix}$$

Since each component of our result is strictly greater than b every inequality constraint is inactive.

Computing all $(a_i^T p)$ or Ap to see which constraints bound our step,

$$\begin{bmatrix} 9 & 4 & 1 & 9 & -7 \\ 6 & -7 & 8 & -4 & -6 \\ 1 & 6 & 3 & -7 & 6 \end{bmatrix} \begin{pmatrix} 3 \\ 2 \\ 0 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 58 \\ 12 \\ -4 \end{pmatrix}$$

Since a_3 is the only constraint which our direction is set to intersect, it defines the $\bar{\alpha}$ in the ratio test,

$$\bar{\alpha} = ((1, 6, 3, -7, 6)(7, -4, -3, -3, 3)^T + 20)/4 = \frac{33}{4}.$$

□

(iii) $x = (5, 0, -6, -8, -3)^T$ and $p = (5, 0, 5, 1, 3)^T$

Solution. Evaluating Ax to see which constraints are active/inactive,

$$\begin{bmatrix} 9 & 4 & 1 & 9 & -7 \\ 6 & -7 & 8 & -4 & -6 \\ 1 & 6 & 3 & -7 & 6 \end{bmatrix} \begin{pmatrix} 5 \\ 0 \\ -6 \\ -8 \\ -3 \end{pmatrix} = \begin{pmatrix} -12 \\ 32 \\ 25 \end{pmatrix}$$

Since each component of our result is strictly greater than b every inequality constraint is inactive.

Computing all $(a_i^T p)$ or Ap to see which constraints bound our step,

$$\begin{bmatrix} 9 & 4 & 1 & 9 & -7 \\ 6 & -7 & 8 & -4 & -6 \\ 1 & 6 & 3 & -7 & 6 \end{bmatrix} \begin{pmatrix} 5 \\ 0 \\ 5 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 38 \\ 48 \\ 31 \end{pmatrix}$$

Since $(a_i^T p) \geq 0$ for all constraints a_i , $\bar{\alpha}$ is unbounded.

□

(iv) $x = (9, 1, -1, 6, 3)^T$ and $p = (-4, -2, 4, -2, 2)^T$

Solution. Evaluating Ax to see which constraints are active/inactive,

$$\begin{bmatrix} 9 & 4 & 1 & 9 & -7 \\ 6 & -7 & 8 & -4 & -6 \\ 1 & 6 & 3 & -7 & 6 \end{bmatrix} \begin{pmatrix} 9 \\ 1 \\ -1 \\ 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 117 \\ -3 \\ -12 \end{pmatrix}$$

Since each component of our result is strictly greater than b every inequality constraint is inactive.

Computing all $(a_i^T p)$ or Ap to see which constraints bound our step,

$$\begin{bmatrix} 9 & 4 & 1 & 9 & -7 \\ 6 & -7 & 8 & -4 & -6 \\ 1 & 6 & 3 & -7 & 6 \end{bmatrix} \begin{pmatrix} -4 \\ -2 \\ 4 \\ -2 \\ 2 \end{pmatrix} = \begin{pmatrix} -72 \\ 18 \\ 22 \end{pmatrix}$$

Since a_1 is the only constraint which our direction is set to intersect, it defines the $\bar{\alpha}$ in the ratio test,

$$\bar{\alpha} = ((9, 4, 1, 9, -7)(9, 1, -1, 6, 2)^T + 15)/72 = \frac{139}{72}.$$

□

Exercise P8: Consider the one-dimensional unconstrained optimization problems,

$$\min_{x \in \mathbb{R}} f(x)$$

for objective functions f which have one continuous derivative f' . I propose the following algorithm

(a) Implement Algorithm P8A. In Matlab it will be a function,

$$\text{function } z = \text{p8a}(f, df)$$

where the inputs are functions $f = f$ and $f' = df$. The output z is the proposed x -coordinate of the solution, i.e. it is the minimizer.

Solution. Consider the following,

Code:

```
function z = p8a(f, df)
% This function takes a function f (doesn't really need it)
% and it's derivative and attempts to find a local
% minimum around with initial point x = 1 with a set
% step size alphak.
x = 1;
alphak = .01;
while true
    %Stopping Criteria.
    if abs(df(x)) < 1e-3
```

```

        break;
    else
        %Descent direction finder.
        if df(x) > 0
            pk = -1;
        else
            pk = 1;
        end
        %Applying descent direction and iterating.
        x = x + alphak*pk;
    end

end

z = x;

```

□

(b) Run the algorithm, and state briefly what happens, in the following cases:

(i) $f(x) = x^2 - 3x + 2$

Solution. Running the algorithm on the given $f(x)$

Console:

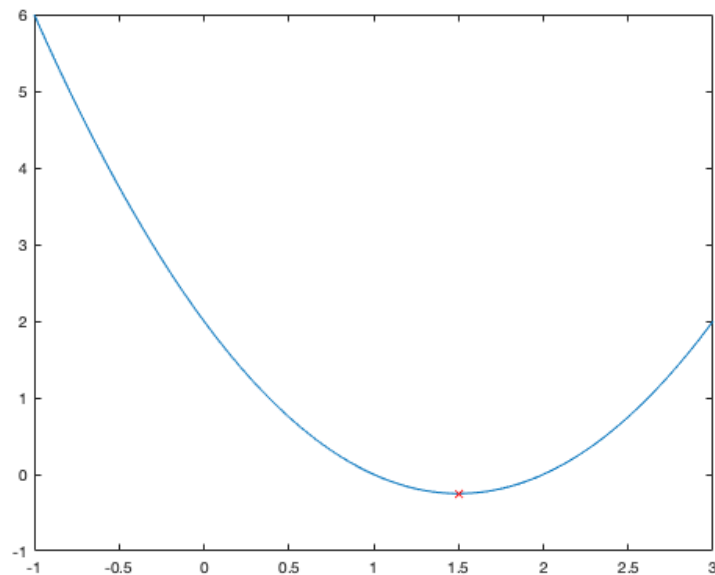
```

>> f = @(x) x.^2 - 3.*x + 2
>> df = @(x) 2.*x - 3
>> z = p8a(f, df)

z =
    1.5000000000000000

>> x = linspace(-1, 3, 100);
>> plot(x, f(x))
>> hold on
>> plot(z, f(z), marker = 'X', color = 'r')

```

Figure 1: Plotting $f(x)$ with minimum from p8a.

□

In this case the algorithm is working in a convex interval, and it starts off at $x_0 = 1$ so we are to the left of the local minimum (in this case it is a global one). Every iteration, the computation of the derivative comes out negative so α_k is always positive so in each iteration x increases by .01 approaching the minimizer from the left.

(ii) $f(x) = \cos(x/50)$

Solution. Running the algorithm on the given $f(x)$

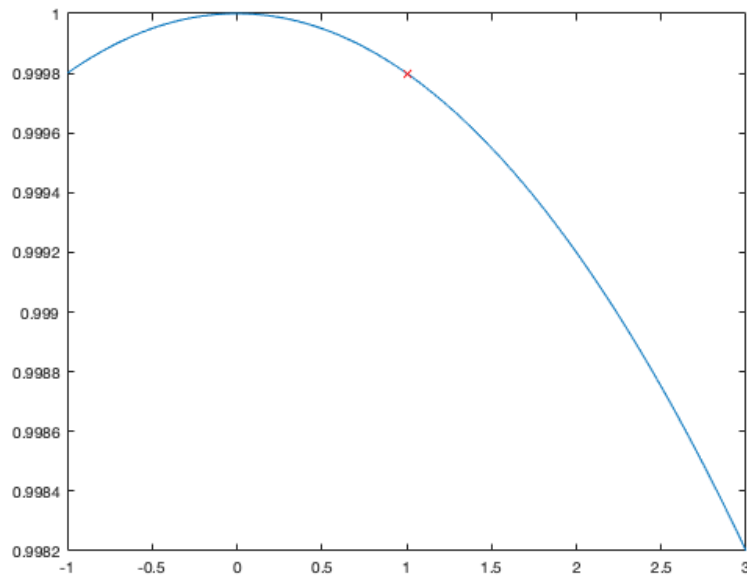
Console:

```
>> f = @(x) cos(x./50)
>> df = @(x) -sin(x./50).*(1/50)
>> z = p8a(f, df)
z =
    1

>> x = linspace(-1, 3, 100);
>> plot(x, f(x))
>> hold on
>> plot(z, f(z), marker = 'X', color = 'r')
```

```
>> df(1)
ans =

-3.999733338666616e-04
```

Figure 2: Plotting $f(x)$ with minimum from p8a.

Clearly a local minimum will be found to the right of the initial $x_0 = 1$. However since the $df(x_0) < .001$ the algorithm stops on the first iteration and returns the initial point as the minimizer. $\square$

(iii) $f(x) = e^{\sin(10x)}$

Solution. Running the algorithm on the given $f(x)$

Console:

```
>> f = @(x) exp(sin(10.*x))

>> df = @(x) f(x).*cos(10.*x).*10

%% Modified to disp(x) after each iteration
>> z = p8a(f,df)
1.1000000000000000

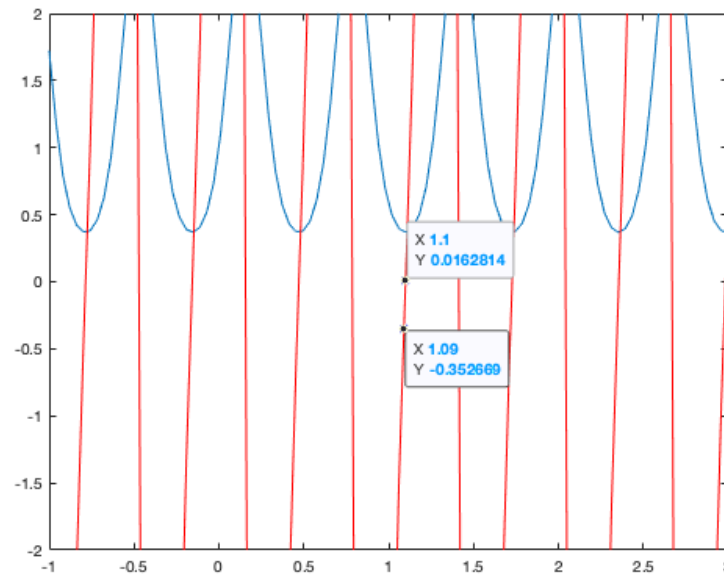
1.0900000000000000

1.1000000000000000
```

1.0900000000000000

The algorithm fails to converge, because the step size α_k is too large we end up hopping back and forth around the minimizer. The following plot illustrates this point, note that $f(x)$ is plotted in blue, $f'(x)$ is plotted in red and the two points, $x_1 = 1.1$ and $x_2 = 1.09$ are plotted on $f'(x)$ with blue crosses.

Figure 3: Plotting $f(x)$, $f'(x)$ and problematic x_1 and x_2 .



□

(iv) $f(x) = \text{sech}(x)$

Solution. Running the algorithm on the given $f(x)$

Console:

```
>> f = @(x) sech(x)

>> df = @(x) (-1).*f(x).*tanh(x)

>> z = p8a(f, df)
z =
    7.6099999999999881

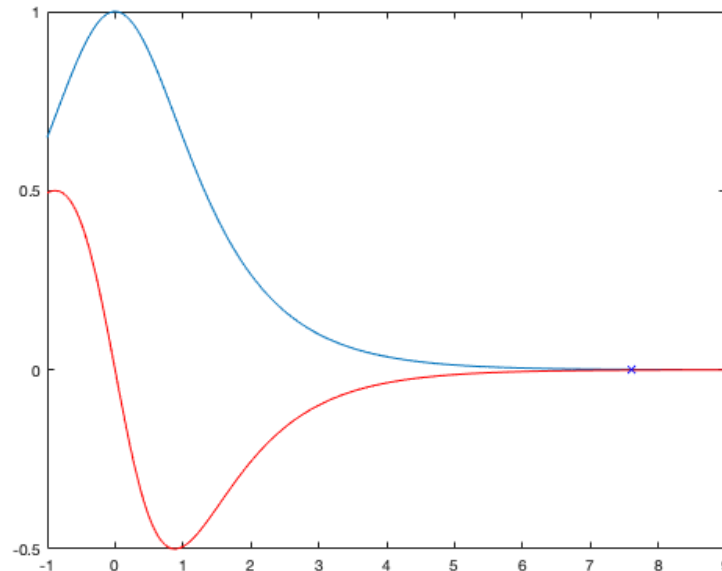
>> x = linspace(-1, 9, 1000);
```

```

>> plot(x, f(x))
>> hold on
>> plot(x, df(x), color = 'r')
>> plot(z, df(z), color = 'b', marker = 'X')
>> plot(z, f(z), color = 'b', marker = 'X')

```

Figure 4: Plotting $f(x)$ with minimum from p8a.



The hyperbolic secant function does not have a global minimum, so this algorithm will continue to add .01 to the current iterant until the derivative becomes a value smaller than .001.

□

- (c) In several complete sentences, describe the main deficiencies of Algorithm P8A. It is a good idea to use the results of part **b** to illustrate some of your points.

Solution. The main criticisms of this algorithm come from the relatively large stopping criteria, and the fixed step sizes and starting point. Consider **b(ii)**, with a stopping criteria of .001 on the derivative the algorithm couldn't even continue to a second iteration, when clearly if dropped into $f(x) = \cos(x/50)$ at $x_0 = 1$ there is a local minimum to the right. Now consider **b(iii)**, the algorithm was unable to terminate because α_k was too large.

□

- (d) *Solution.* The choice of having $|f'(x_k)| < \epsilon_{\text{machine}}$ is the correct one. Any comparison to zero should be made with $\epsilon_{\text{machine}}$ instead because of the limits of our double precision arithmetic. Being able to decide on an initial point, will allow us to at least have some say on the runtime, and which local minimizer we converge to. Finally being able to choose a step size will help us avoid situations illustrated in **b(iii)** where the step size was too small, we get to make the tradeoff between accuracy to the true minimizer and runtime.

This algorithm assumes that the user has some idea about the location of the minimizer on $f(x)$ hence the initial value parameter. It also assumes that the user has some idea of the concavity of $f(x)$, choosing the step size too large for highly concave or convex function will result in a situation like **b(iii)**.

□